Muhammad Memon

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Portfolio: Engineering Portfolio. Web
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Personal Statement

- Hardworking student studying Computer Engineering programme at TU Dublin.
- Developing cloud infrastructure, network configuration, database management, data analytics and eager to further my knowledge and contribute to innovative IT and system development.
- My work experience has given me the opportunity to develop skills in customer service, teamwork and time management.
- Computer Engineering student with practical experience in AI/computer vision, embedded hardware integration, IoT systems and software development. Strong collaborator with proven leadership experience in team projects and volunteering.

Skills Profile

- Programming skills: C/C++, Python, Linux, Grafana/SQL databases, Java/JavaScript
- Network Technologies • Office 365 and systems administration skills • Effective team management • Customer service skills

Academic & Professional Qualifications

2015–2022 BSc (Hons) Computer Engineering
Technological University Dublin, Blanchardstown, Dublin 15
Modules: Computer Networks, Networking, Data Analytics, Java & Algorithms, Embedded System

Projects

AI ADAS Detection System (4th Year Final Project):

- Developed a low-cost driver-assistance prototype using YOLOv8 and OpenCV to detect vehicles and pedestrians on embedded Raspberry Pi hardware.
- Integrated an HC-SR04 ultrasonic sensor for distance measurement, with collision-warning thresholds and a Node-RED dashboard using WebSocket communication.

GP Virtual Monitoring System (3rd Year Final Year Project):

- Developed a health monitoring platform tracking temperature, blood pressure, and heart rate using DS18B20 and SEN0203 sensors, Azure integration and MySQL (XAMPP) for secure storage.
- Designed a dynamic website (HTML, CSS, JavaScript, PHP) for real-time health data access.

Networked Microcontroller IoT System (3rd Year Mini Project):

- Built an ESP8266-based microcontroller with Tasmota firmware for I²C/SPI communication; integrated Microsoft Azure, Node-RED, Grafana and InfluxDB for cloud data management and real-time monitoring.

Greenhouse Monitoring System (2nd Year):

- Designed a real-time system to track temperature, humidity and light (DS1820, DHT22, LDR) with LCD and PCB layout optimisation; tested sensor accuracy and system performance.

Work Experience

Jun 2020 – Jul 2021 Memon Transport Limited, Summer Analyst, Dublin

- Supported system documentation and onboarding processes for new staff; assisted with monitoring and troubleshooting cloud-hosted servers; participated in strategy meetings with senior leadership.

Jul 2022 – Jan 2023 Wow Burger Dublin, Supervisor, Dublin

- Helped customers and reached weekly sales targets; handled customer needs and delivered detailed financial reports to company directors.

Dec 2024 – Dec 2026 Founder, Irish Sign Language Society, TU Dublin

- Established and grew membership to 50+, securing interpreters for inclusivity; promoted Irish culture and inclusivity through Irish and UK Sign Language.

Interests & Achievements

- Working on projects; maintaining computer hardware/software; badminton, cycling and fitness; volunteering as Treasurer for Charity Week and ambassador roles in Amazon & Deliveroo.
- Gaisce President's Award and programming awards through Coursera/Udemy; 3rd Prize at BT Young Scientist of the Year; nominated Class Representative Student Ambassador of the Year – 2024 & 2025.